2007级数学物理方程期末试题（A卷）（答案）

（信二学习部整理）

一.

(1).
$$\begin{cases}\dfrac{\partial u}{\partial t}=a^2\dfrac{\partial^2 u}{\partial x^2} & 0<x<l,\ t>0 \quad \text{------ 2分}\\ u|_{x=0}=u_0,\ \dfrac{\partial u}{\partial x}\Big|_{x=l}=0 & \text{------ 4分}\\ u|_{t=0}=\varphi(x) & \text{------ 6分}\end{cases}$$

(2)
$$\begin{cases}\dfrac{\partial^2 u}{\partial t^2}=a^2\dfrac{\partial^2 u}{\partial x^2} & \text{------ 2分}\\ u|_{x=0}=0,\ u|_{x=l}=0 & \text{------ 4分}\\ u|_{t=0}=\begin{cases}4hx & 0\le x\le\frac{1}{4}\\ \frac{4}{3}h(1-x) & \frac{1}{4}<x\le 1\end{cases}\\ u_t|_{t=0}=0 & \text{------ 6分}\end{cases}$$

(3)
$$\begin{cases}\dfrac{\partial^2 u}{\partial \rho^2}+\dfrac{1}{\rho}\dfrac{\partial u}{\partial \rho}+\dfrac{1}{\rho^2}\dfrac{\partial^2 u}{\partial \theta^2}=0 & 0<\rho<a \quad \text{------ 4分}\\ u|_{\rho=a}=f(\theta) & \text{------ 6分}\end{cases}$$

二. 令 $u(x,t)=v(x,t)+w(x)$ 其中 $w(x)$ 满足

$$\begin{cases}a^2w''+A=0\\ w(0)=0,\ w(l)=B\end{cases} \quad \text{------ 3分}$$

即 $w(x)=-\dfrac{A}{2a^2}x^2+\left(\dfrac{B}{l}+\dfrac{Al}{2a^2}\right)x$ ------ 5分

则 v 满足:

$$\begin{cases}\dfrac{\partial^2 v}{\partial t^2}-a^2\dfrac{\partial^2 v}{\partial x^2}=0\\ v|_{x=0}=0,\ v|_{x=l}=0\\ v|_{t=0}=-w(x),\ v_t|_{t=0}=0. & \text{------ 7分}\end{cases}$$

分离变量法 $v(x,t)=X(x)T(t)$ 则

$$\begin{cases} X''+\lambda X=0 \\ X(0)=0,\ X(l)=0 \end{cases}$$

$$T''+\lambda a^2 T=0$$

本征值 $\lambda=\frac{n^2\pi^2}{l^2}$ $n=1,2,\cdots$

本征函数 $X_n(x)=\sin\frac{n\pi x}{l}$ $n=1,2\cdots$ ---- 10分

$$T_n(t)=C_n'\cos\frac{n\pi at}{l}+D_n'\sin\frac{n\pi at}{l}$$ ----- 12分

$$u(x,t)=\sum_{n=1}^{\infty}\left(C_n\cos\frac{n\pi at}{l}+D_n\sin\frac{n\pi at}{l}\right)\sin\frac{n\pi x}{l}$$ ---- 13分

由初始条件 $v_t|_{t=0}=0 \Rightarrow D_n=0$

$$v|_{t=0}=-w(x)\Rightarrow C_n=\frac{2}{l}\int_0^l -w(x)\sin\frac{n\pi x}{l}dx$$

$$C_n=(-1)^n\frac{2B}{n\pi}+((-1)^n-1)\frac{2Al^2}{n^3\pi^3a^2}$$ - - - -

故 $v(x,t)=\sum_{n=1}^{\infty}C_n\cos\frac{n\pi at}{l}\sin\frac{n\pi x}{l}$

$$u(x,t)=v(x,t)-\frac{A}{2a^2}x^2+\left(\frac{B}{l}+\frac{Al}{2a^2}\right)x$$ --- 15分

三. $\begin{cases} \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0 \quad 0<x<a. \quad 0<y<b. \\ u|_{x=0} = u|_{x=a} = 0 \\ u|_{y=0} = \sin\dfrac{\pi x}{a}. \quad u|_{y=b} = 0 \end{cases}$

解, 分离变量. $u = X(x)Y(y)$ 得. $\dfrac{X''}{X} = -\dfrac{Y''}{Y} = -\lambda$

由边界条件, $X(0) = X(a) = 0$ 则特征值问题:

$$\begin{cases} X'' + \lambda X = 0 \\ X(0) = X(a) = 0 \end{cases}$$

特征值. 特征函数.

$$\lambda_n = \frac{n^2\pi^2}{a^2}. \quad X_n = C_n \sin\frac{n\pi}{a}x, \quad n = 1.2\cdots \quad \cdots\cdots 5$$

代入 $Y'' - \lambda Y = 0$. 得.

$$Y_n = A_n' e^{\frac{n\pi}{a}y} + B_n e^{-\frac{n\pi}{a}y} \quad \cdots\cdots 8$$

则得. $u_n = X_n Y_n = \left(A_n e^{\frac{n\pi}{a}y} + B_n e^{-\frac{n\pi}{a}y}\right)\sin\frac{n\pi}{a}x \cdots\cdots 10$

由叠加原理.

$$u = \sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty}\left(A_n e^{\frac{n\pi}{a}y} + B_n e^{-\frac{n\pi}{a}y}\right)\sin\frac{n\pi}{a}x \cdots\cdots 12$$

由 $u|_{y=0} = \sin\frac{\pi x}{a} \quad u|_{y=b} = 0$. 得.

$$\begin{cases} \sum_{n=1}^{\infty}(A_n + B_n)\sin\frac{n\pi}{a}x = \sin\frac{\pi x}{a} \cdots\cdots ① \\ \sum\left(A_n e^{\frac{n\pi b}{a}} + B_n e^{-\frac{n\pi b}{a}}\right)\sin\frac{n\pi}{a}x = 0 \cdots\cdots ② \end{cases}$$

由①. 得: $A_1 + B_1 = 1$. $\quad A_n + B_n = 0 \quad n = 2,3,\cdots$

由②. 得. $A_n e^{\frac{n\pi b}{a}} + B_n e^{-\frac{n\pi b}{a}} = 0. \quad n = 1.2\cdots$

于是：$A_1 = \left(1 - e^{\frac{2n\pi b}{a}}\right)^{-1}$

$$B_1 = 1 - \left(1 - e^{\frac{2n\pi b}{a}}\right)^{-1}$$

$$A_n = B_n = 0 \quad n = 2, 3, \cdots$$

则：$u(x,y) = \dfrac{1}{1 - e^{\frac{2n\pi b}{a}}}\left[e^{\frac{n\pi}{a}y} - e^{\frac{2n\pi b}{a} - \frac{n\pi}{a}y}\right] \sin\dfrac{\pi x}{a}$

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四：$\begin{cases} 3u_{tt} - 2u_{tx} - u_{xx} = 0 & t>0,\ -\infty < x < \infty \\ u|_{t=0} = 3x^2 \\ u_t|_{t=0} = 0 \end{cases}$

解：是双曲型偏微分方程.

特征方程：

$$3(dx)^2 + 2\,dx\,dt - (dt)^2 = 0$$

即：$3\left(\frac{dx}{dt}\right)^2 + 2\frac{dx}{dt} - 1 = 0$

特征曲线：$\begin{cases} 3x - t = C_1 \\ x + t = C_2 \end{cases}$

特征变换，$\begin{cases} 3x - t = \xi \\ x + t = \eta \end{cases}$ 4

经特征变换，方程化为：$\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$ 6

通解：$u = f_1(\xi) + f_2(\eta)$ 8

其中 f_1、f_2 是任意二次可微连续函数.

则：$u(x,t) = f_1(3x - t) + f_2(x + t)$

由边界条件：$\begin{cases} f_1(3x) + f_2(x) = 3x^2 \\ -f_1'(3x) + f_2'(x) = 0 \end{cases}$

第二式积分：

$$-\frac{1}{3}f_1(3x)+f_2(x)=C$$

$$\Rightarrow\begin{cases} f_1(3x)=\frac{9}{4}x^2-\frac{3}{4}C \Rightarrow f_1(x)=\frac{x^2}{4}-\frac{3}{4}C. \\ f_2(x)=\frac{3}{4}x^2+\frac{3}{4}C \end{cases}$$

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则：

$$u(x,t)=f_1(3x-t)+f_2(x+t)$$

$$=\frac{(3x-t)^2}{4}+\frac{3}{4}(x+t)^2.$$

$$=3x^2+t^2.$$

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五、解 对x进行 Laplace 变换

$$\begin{cases} \frac{\partial}{\partial y}(pU-y^2)=\frac{6}{p^3} \\ U|_{y=1}=\frac{1}{p} \end{cases}$$

解得 $U(p,y)=\frac{6y}{p^4}+\frac{y^2}{p}-\frac{6}{p^4}$

反变换得

$$u(x,y)=x^3y+y^2-x^3$$

六 (1) 取 $M_0(x_0, y_0)$ 像点 $M_1(4-x_0, y_0)$

所以 Green 函数为 $G(M, M_0) = \frac{1}{2\pi}\left[\ln\frac{1}{r_{MM_0}} - \ln\frac{1}{r_{MM_1}}\right]$

$$= \frac{1}{2\pi}\ln\frac{\sqrt{(x-(4-x_0))^2+(y-y_0)^2}}{\sqrt{(x-x_0)^2+(y-y_0)^2}}$$

(2) $\left.\frac{\partial G}{\partial n}\right|_{x=2} = -\left.\frac{\partial G}{\partial x}\right|_{x=2} = -\frac{1}{\pi}\frac{2-x_0}{(x-x_0)^2+(y-y_0)^2}$

所以解为 $u(x_0) = -\int_{x=2}\frac{\partial G}{\partial n}\varphi(y)\,ds = \frac{1}{\pi}\int_{-\infty}^{+\infty}\frac{2-x_0}{(x-x_0)^2+(y-y_0)^2}\varphi(y)\,dy$